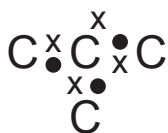


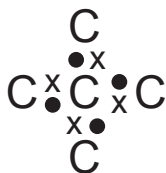
7. Carbon exists in many forms. Each carbon atom has 4 electrons available for bonding.

(a) Four different dot and cross diagrams are shown below.

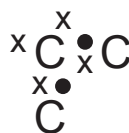
A



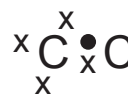
B



C



D



(i) State which diagram shows the bonding found in diamond. [1]

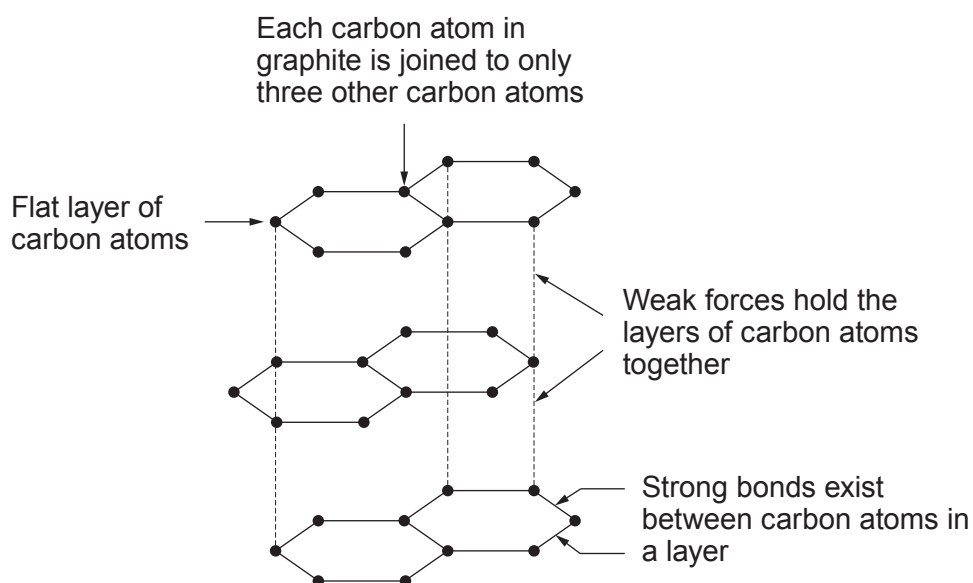
(ii) Give a reason for your choice. [1]

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(iii) Name the type of bond found between carbon atoms in diamond. [1]

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- (b) Carbon also exists in the form of graphite. The diagram shows the giant molecular structure of graphite.



- (i) Explain why graphite conducts electricity.

[2]

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- (ii) Explain why graphite is slippery.

[2]

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